Inertia and Transformation

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Abstract

Strategy scholars have incorrectly borrowed from economists the assumption of organizational plasticity. Particularly in large firms, inertia, rather than plasticity, is the norm. Unfortunately, there can be no simple theory of inertia as its causes are multiple and varied. After sketching out the shapes of the most important sources of inertia, I turn to the problem of overcoming inertia-the question of organizational transformation. Starting with a simple model of organizational capabilities as existing on two levels (unit-based and rooted in coordination among units), I draw some preliminary conclusions about the shape of organizational transformation. In particular, I focus on the interplay between incentive intensity and coordinative capacity and argue that most transformations move through a sequence of phases in which coordinative capacity is first dramatically reduced and then rebuilt along new lines.

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Introduction

Roughly fifteen years ago the field of business and corporate strategy began to incorporate economic reasoning into its research program. The first step was the adoption of traditional industrial organization economics, with its emphasis on barriers to entry and collusive reductions in rivalry (Porter 1980). Subsequently, strategy researchers developed what is now called the resource-based view of the firm. [2] This theory sees firms as collections of resources and sees performance differences as largely reflecting differences in resource quality. Whereas traditional industrial organization saw high profits as stemming from collusive reductions in competition or strategies of entry deterrence, the resource-based view sees high profits as the rents accruing to specialized and difficult-to-replicate or non-imitable resources.

Thus, today strategy researchers work with a complex amalgam of economic and quasi-economic reasoning. We envision the firm as striving to maximize value, but also see it as working with factors

[1] Financial support from INSEAD's Corporate Renewal Initiative (CORE) is gratefully acknowledged. My understanding of the issues surrounding corporate transformation has benefitted from discussions with Sumantra Ghoshal, Chris Bartlett, and John Stopford.

of production that are far from mobile, as dealing with ambiguous production functions, and as possessing or controlling collections of tacit knowledge and externally held attributions (reputation) that evolve over time in response to investment, activity and imitation.

It is useful to note that this new view has not been developed by "applying" economics to strategy. Rather, it has been accomplished by carefully identifying the assumptions within received economic models that prevented or ruled out strategic phenomena, and by then analyzing the situations created by altered assumptions. In particular, a central dogma of neoclassical microtheory, especially that part associated with traditional industrial organization, was the basic homogeneity of firms within industries (but for scale). Differences in performance were thus attributed to differences in scale or its collective equivalent, concentration. By contrast, the resource-based view has as its central dogma the heterogeneity of firms induced by heterogeneous resources.

The power behind this research stream has, in large measure, come from the clarity and strength of the theory (neoclassical microeconomics) which it attacks. In fact, the neoclassical model maintained such a grip on the minds of economists that they often spoke of its failures as "market failures," as if it were somehow the responsibility of reality to live up to theory rather than the theorist's responsibility to describe reality. One problem with this research strategy is that the foil, neoclassical microtheory, has died. Game theory now provides the basis for reasoning in industrial organization economics; agency theory provides explanations for non-optimal firm behavior, and transaction-cost economics describes the limits of markets in carrying out coordination and exchange. One might argue that this is fine-the resource-based view together with revived microtheory is the strategic theory of the firm. However, I believe that despite the flowering of this rich new microtheory, we have dealt with only one part of the story. There are a number of erroneous assumptions that most economically-oriented strategy researchers continue to borrow from economics. At this moment those that are clearest are plasticity, rationality of collective action, and homogeneity of beliefs. I believe that the most important of these is plasticity-the assumption that firms readily respond to exogenous shocks and changes in competitive conditions. The centerpiece of microeconomics is the deduction of the total economy's autonomous responsiveness (mediated by self-interest) to changes in prices, technology, taxes, etc. Yet the truth is that firms change only with difficulty. Changing strategy and the structural forms and administrative procedures that undergird strategy is difficult, costly, risky, and time consuming.

I shall call this lack of plasticity inertia. Inertia is the strong persistence of existing form and function. If the form is efficient, inertia is costless and arguably beneficial. However, if the firm's form or practices are inefficient, inertia is a problem. Indeed, the most direct evidence of inertia is the persistence of inefficient forms and practices. [3]

A widely cited example of organizational inertia is General Motors. Once the world leader in

[2] Early contributions were made by Lippman and Rumelt (1982), Teece (1982), Wernerfelt (1984), Rumelt (1984), Porter (1985) (who made the "activity" the central element of his revised view), and Barney (1986). Reviews of the topic are provided by Conner (1991) and Grant (1991).

automobile production efficiency, General Motors was eclipsed in this regard by leading Japanese manufacturers during the 1970s. Senior management understood the nature of the productivity gap by 1979, yet, despite a joint venture with Toyota in which world-class methods were used, the company has been unable to change its overall productivity in any substantial way. In fact, many GM plants became less productive during the 1980s, while Chrysler and Ford made broad and significant gains. Clearly, the basic problem facing the senior management of General Motors is not product-market strategy, but organizational change. Their challenge is not really competition, but their company's own inertia.

The fact of organizational inertia is not simply an "implementation" problem. If firms lack plasticity then the formulation of product-market and corporate strategy is itself fundamentally altered:

Good product-market strategy must take into account a firm's inertia and not create new inertia without sufficient reason.

The important strategy problem facing a firm may well be internal inertia rather than product-market conditions.

Leaving inertia out of an analysis underestimates, perhaps drastically, the payoff to strategic change and innovation. Modern economic analysis of strategy presumes alert responsive competitors who will compete away a firm's profits unless there are the protections afforded by property rights, reputation, tacit know-how, or other strategic resources and isolating mechanisms. But if competitors are subject to inertia, this analysis is incorrect. Some of the great strategic success stories are due as much to the (temporary) inertia of competitors as to the cleverness of the innovator (e.g., Timex, Federal Express).

A basic question motivating much of the research into the resource-based point of view was "Why are firms different?" The question I suggest requires equal attention is "What are the sources of organizational inertia?" Or, "Why is change so difficult?"

Inertia and Evolution

The inertia of firms is a dominant theme in two fields loosely coupled with strategy: organizational ecology and evolutionary economics. Main-line organizational ecology assumes that firms are immutable (once born) and that populations of firms change over time through processes of birth, selection, and death. In evolutionary economics there is the additional theme that firms change, though not always adaptively, as occasional innovation and trial-and-error processes affect their work-routines and policies.

Within organization ecology, Hannan and Freeman (1984) define inertia as change that is slow

[3] One research stream in economics labels persistent inefficiency in the use of inputs as "X-inefficiency," in contrast to the more commonly understood problem of allocative inefficiency. For a survey of this literature as well as empirical estimates of the amount of X- inefficiency see Frantz (1988)].

relative to the environment, so that a firm that is considered flexible in one setting might still have too much inertia to compete in a still faster-changing industry. Hannan and Freeman argue that much inertia is the unavoidable consequence of specialized investment in physical assets and social structures. In particular, they argue that these specialized investments meet the organizations need for reliability and accountability-the need to repeatedly produce the same type of product or process and the need to justify and explain process and choice to outsiders. They identify reproducibility with institutionalization, which gives "an organization a taken-for-granted character such that members do not continually question organizational purposes, authority, relations, etc." (1984:154). Hannan and Freeman follow other sociologists in arguing that organizations have "cores" which are very difficult to change relative to more peripheral elements. The four core elements they identify are: stated goals, the form of authority, the basic technology, and the marketing strategy.

Within evolutionary economics the explanation for inertia is rooted in bounded rationality, routines, and tacitness. Nelson and Winter (1982) argue that the skills and capabilities of organizations are bound up in their routines for accomplishing tasks. These routines, and the fact that they are routine, constitute the skill set and the memory of the firm. Within this framework, inertia is the natural state of affairs-firms can only do what they have routines for doing. Lacking a routine for a new task, the new task does not really get done.

These perspectives are both helpful in understanding inertia. However, neither attempts to be comprehensive or analytic. In the next sections I begin to build a framework that attempts to encompass a broader managerial view of inertia and begins to allow analysis of the constellation of inertial forces at work in different situations.

The Five Frictions

With a nod to Michael Porter, I have organized the main sources of inertia into five groups, called the five frictions:

Distorted Perception

Dulled Motivation

Failed Creative Response

Political Deadlocks

Action Disconnects

In many cases, the components of these frictions are well known or easily comprehended. In other cases the issues are novel or subtle. I shall pass swiftly over many the straightforward issues in order to concentrate on those that are novel or complex.

Distorted Perception

Change begins with perception. If perception is distorted, then change may be impeded. The fundamental sources of perceptual distortion inducing organizational inertia are: myopia, hubris and denial, and grooved thinking.

Myopia. A firm suffering from myopia is unable to look into the future with clarity. Various individuals within the firm may be aware of the future consequences of current action or inaction, but the organization, taken as a whole, acts as if only the short-term matters.

The simplest source of myopia is turnover. If a manager expects to move to another firm in the near future, the weight placed on future profits is diminished. Suppose, for example, that the natural annual discount rate is r, that a manager believes the chance of leaving in any given month is constant (i.e., an exponential departure time) and believes the expected time to departure to be τ years. Then the manager's effective discount rate [4] for decision-making is $r + 1/\tau$. Thus, the 20 percent annual turnover in managers experienced during the halcyon years of Silicon Valley may have increased the effective discount rates used in decision-making from a normal 15 percent to 35 percent. Myopia is to be expected in a firm that has employed high-intensity short-term controls. In such a case individuals bias their allocations of attention and effort toward problems that effect current performance. Less obvious is the fact that myopia is characteristic of a firm which is "forgetful." That is, when individuals do not expect the organization to remember the connections between current actions and future results. In forgetful firms, characterized by mergers, reorganizations, frequent personnel changes, etc., individuals have no realistic expectation that future results will be attributed to their current actions or decisions.

When managers are myopic, the subtle consequence is that senior management will also behave myopically, expressing a rational disbelief in the claims of future threat or opportunity expressed by lower level managers. Thus the expectation of myopia in others leads, in turn, to further myopic behavior.

A final source may be the failure of planning systems to counteract normal human myopia. Empirical evidence suggests that people's time-discount functions are steeper-than-exponential, leading to timeinconsistent behavior. [5] Well-functioning organizational systems, it may be argued, help counteract short-term impulsiveness. And, by contrast, poorly functioning planning and review systems fail in this regard, fostering organizational myopia.

Hubris and denial. A serious source of perceptual distortion is denial-the rejection of information that is contrary to what is desired or what is believed to be true. Denial may stem from hubris-overweening pride in past accomplishments-or it may derive from fear. In his work on barriers to organizational learning, Argyris (1990) has argued that a virtually universal organizational response to information or analysis which threaten a loss of face is defensive behavior: denial of the validity of the data, cover-up of the situation, and cover-up of the cover-up. For example, for years the

[4] This analysis assumes no ex-post settling up. That is, when leaving the firm the manager escapes the future negative consequences of myopic decisions.

senior management of General Motors refused to admit that Japanese firms had higher absolute levels of productivity, denying the existence of the problem. In part, management could not believe that another company knew more about producing cars than did GM (hubris), and in part it was less embarrassing to believe other theories about the rising tide of imports.

Hubris is also explainable as superstitious learning-learning based on associating past success with factors that were coincidental with it but bear no causal relationship to the success. Skinner (1948) showed that when pigeons receive random reinforcements (feedings), they become conditioned to repeat some behavior that is accidentally correlated with the stimulus. Similarly, managers in highly uncertain environments (e.g., the entertainment industry) may cling steadfastly to policies that were only accidentally correlated with some past success.

A related syndrome is information filtering-the selective rejection of information that is unpopular, unpleasant, or contrary to doctrine. During the Vietnam war, for example, U.S. commanders defined the conflict as a war of attrition which was being won over time and systematically withheld from Washington information which contradicted their vision (Sheehan 1988).

Grooved thinking. Janus (1972) has described as "groupthink" the restricted thinking that groups impose, punishing or rejecting ideas and information that deviate too much from orthodoxy. A somewhat different perspective is provided by Margolis (1993) who views patterns of thinking as mental habits whose structure and function are the same as physical habits. Finally, a third type of grooved thinking comes from the use of the "wrong" metaphor. Just as policy makers may struggle to decide whether the situation in Bosnia is "another Kuwait" or "another Vietnam," a metaphor, once accepted, acts as a powerful restriction on future thought.

Dulled Motivation

Even if perception is accurate, organizations may resist change because the need is not felt with sufficient sharpness. The lack of sufficient motivation may be rational, or it may reflect agency or psychological problems. The most important motivational dampers are: direct costs of change, cannibalization costs, and cross-subsidy comforts.

Direct costs of change. It is likely that change temporarily increases the risk of organizational failure (mortality), disrupts operations, and involves a great deal of expensive effort. Even more importantly, change may imply the abandonment of costly sunk specific investments. If these considerations apply to the firm as a whole, they are rational impediments to change. Applied to individuals or groups, they point to agency problems.

Note that an impediment to change may be rational. When Timex created the disposable watch, there were no imitators for ten years. Some of the inertia was surely wasteful, but some may have been rational. Were a quality Swiss manufacturer to produce disposable watches it would risk damage to its reputation. Similarly, whereas firms making new investments may be able to justify

[5] See Postrel and Rumelt (1992) for a full discussion.

an expensive new technology, a firm with adequate though less efficient capacity may not be able to economically justify switching to the newer technology.

Cannibalization costs. When a new product's success eats into the sales and profits of an older product, the older product is said to have been cannibalized. Cannibalization problems may be rational or simply reflect sub-group interests. Rational cannibalization problems occur under conditions of buyer loyalty (or switching costs). Loyal buyers will stay with a firm's old product despite competitors' introductions of new versions, but will switch to the new version when offered by the firm they favor.

This asymmetry in buyer response, when coupled with the fact of lower profits on the newer products, can induce a firm to exploit its buyers' loyalties by withholding introduction of the new product. For example, when interest rates rose sharply in the late 1970s, many banks and S&Ls began to offer new money-market accounts and other innovative services. However, those institutions with the largest volume of savings account deposits (paying 5 percent interest when market rates were 17 percent) were least likely to market the new products. They did, of course, lose some customers to other banks. But they also reasoned that they would lose even more savings account customers were they to market the new products heavily themselves-the new products would cannibalize the very high profits being earned on the old products.

Cross subsidy comforts. The motivation to change is inhibited when a problem business is subsidized by rents from another business. The subsidy may be direct, in the form of management's toleration of losses in a business that is compensated for by gains elsewhere. Or, it may be indirect, obtained through artificial transfer prices or through bundling businesses together so that separate measures are not obtained.

Failed Creative Response

If perception is acute and motivation sharp, change may still be blocked by other forces. In particular, it may be difficult for the organization to choose a direction out of its difficulties. The impediment may be in the analysis of the situation or in choice itself. The major categories of friction in this area are: speed and complexity, reactive mind-set, and inadequate strategic vision.

Speed and complexity. In the 1970s John Boyd, a captain in the U.S. Air Force, concluded a study of why U.S. aviators had been so successful in air combat in Korea. [6] He found that although the MiG-15 was faster, quicker in the turn, and could climb faster than the U.S. F-86, the critical factor was that the F-86 provided better visibility and faster control. Thus the U.S. pilot could remain a step ahead, maintaining the initiative. Most importantly, Boyd found that once the MiG pilot found himself outmaneuvered, he froze or panicked.

Analysis is blocked or frozen when things happen too fast. If a competitor can go around the "Boyd loop" of observation, orientation, decision, and action faster than an opponent, the opponent may not simply struggle along, he may freeze-up or collapse. The "Boyd loop" phenomena is one

form of what is known to everyone as "having the initiative" in a game.

When the decision situation is very complex, there may also be a similar blockage. For example, U.S. firms all dropped out of the liquid crystal flat-screen technology race when Japanese firms jumped too far ahead. The pace of events overtook their resource allocation systems and the complexity of the judgments to be made exacerbated the situation.

Reactive mind-set. Change is inhibited when people adhere to the view that their problems are natural and inevitable. The most common reactive mind-sets are that the industry is "mature" that the problems are industry problems and not the fault of the firm. These points of view have great validity behind them-the weight of expert advice and analysis. They are also self-fulfilling. If all competitors define their market as mature they will surely be correct.

Senior management may also be stuck in a reactive mind-set if they possess too little detailed understanding of the business to take a proactive creative stance. Entrepreneurial creativity requires a closeness to issues of commerce, technology, and/or buyer behavior that may be lacking in a senior management group that came to power in a different era or who have backgrounds unconnected to current business issues.

Inadequate strategic vision. Even when analysis and choice have not been blocked, the direction chosen and especially its articulation may be so flawed that change is blocked. Vision (direction) also may be inadequate because it is hypocritical. Hypocritical vision is dishonest, claiming values and goals that are known to be false. Announcing "We are a community ..." is a lie for a firm that is about to lay off one-third of its workforce. Managers know when goals of "quality" or of being "ecologically sound" are hypocritical and cannot help but treat the rest of the vision with cynicism.

Vision can be inadequate because it is unresponsive to the clear challenges facing the firm. An unresponsive vision can hardly inspire commitment and change. Finally, vision can be ineffective because there is no trust in senior management's commitment to the vision. If people believe that this "vision" is simply today's plan, only to be replaced by another tomorrow, there will be no willing followers. If managers are to commit their energy, careers, time, and attention to a program of change, there must be trust that the direction chosen will not be lightly altered. Here we touch the central paradox that change may require the promise of future inertia.

Political Deadlocks

Politics, especially organizational politics, is frequently seen as somehow tainted or improper. But politics is simply about influence on decisions and is absent only in absolute dictatorships. The three main sources of disagreement among men are differences in personal interest, differences in belief, and differences in fundamental values. These are also the underlying themes of the three types of political deadlocks: departmental politics, incommensurable beliefs, and vested values.

[6] See Fallows (1981) and Smith (1985). Thompson (1992) provides a businessoriented interpretation.

Departmental politics. This is one of the most obvious sources of inertia and little more need be said. Managers rarely act to unseat themselves or to terminate their own departments. Yet change inevitably involves winners and losers; some people and departments will gain resources and prestige and others will be lessened. Change will be fought by those who will clearly lose thereby and departmental wrangling over who will win and lose can slow change to a crawl.

Incommensurable beliefs. More interesting than the politics of self interest is the problem that arises when different individuals or groups hold sincere but differing beliefs about the nature of the problem or its solution. Figure 1 illustrates the classic Condorcet paradox. Here three managers face three alternatives (downsize, merge, and R&D), and each ranks the three differently. The problem here is that the group's preferences are intransitive. If any of the three alternatives is taken to be the status quo, there is always an alternative preferred to it by a 2 -to- 1 majority. Here "R&D" beats "downsize," "merge" beats "R&D," and "downsize" beats "merge." There is no stable resting place among the cyclic preference mix.

	Manager		
	John	Jan	Jon
Downsize	1	2	3
Merge	2	3	1
R&D	3	1	2

Figure 1 Condorcet Preferences

This sort of problem in aggregating individual preferences led Arrow (1963) to his Impossibility Theorem and has inspired a stream of related work. It all points to the fact that traditional concepts of "public interest" or the "organization's goals" are empty of rigorous meaning unless there is uniformity of opinion.

What happens when these managers are asked to reach a "consensus?" Modern political theory establishes that here (and in general) there is no "rational" outcome, and that any of the outcomes can be obtained by manipulation of the agenda. [7] My belief is that there are four basic patterns that can emerge. First, a leader may simply impose a decision, eliminating the illusion of group choice. Secondly, the choice process may cycle for some time without generating an outcome. Thirdly, the managers may recognize the "irrationality" of the situation and withdraw from participation (essentially colluding to avoid choice). Finally, other considerations or "games" may change the relative influence the managers have, resolving the paradox.

When managers disagree, there is no "rational" way to combine their beliefs. If speed is mportant, leadership may have to abandon group decision processes. When process is important, and beliefs differ, inertia may well be the outcome.

Vested values. The third source of political deadlock is the presence of vested values. Unlike the cases of differing interests or beliefs, here individuals and departments are taken to have strong

emotional or value attachments to products, policies, or ways of doing things. These vested values and interests can easily be the greatest impediments to change.

For many years a central concept in the strategy field was the experience curve-the idea that the cost of doing something falls as cumulative experience (production) increases. By analogy, I suggest that there is a vested value "curve": the more one does a task the more one sees value in that activity. I am careful not to say one "likes" the activity, for that is another issue. [8]

The psychology of vested value is dissonance reduction-the repetition of something difficult makes sense if it is valued. If one has been rewarded for doing the task, it will not only be valued, but liked as well.

Vested values lie at the heart of institutions. The defenders of vested values are usually the informal leadership network-the defenders of the society and of its norms. The paradox of change is that these same people, perhaps the best and the brightest, easily become the source of inertia. The problem of vested values is not with simple foot dragging, but with the organizational equivalent of patriotism.

Action Disconnects

The fifth source of friction concerns those forces which prevent action. Even if perception has been sound, analysis and choice have proceeded, and the problems of politics overcome, there may still be no change. The basic reasons for action blockades and disconnects are: leadership inaction, embedded routines, collective action problems, and capabilities gaps.

Leadership inaction. For change to begin, the leadership must articulate a vision for change, must alter incentives, must take direct action where possible, and must shift power. If it fails to do these things, change will be inhibited. The need for direct action is particularly worth emphasizing. The concept of leadership comes from the military notion of being "in front," of providing courage and energy by example. In business, the analog is for the senior management group to lead by example. If a new emphasis on service is the centerpiece of the strategy, and the marketing manager keeps on an old friend, the head of market research, despite a wide reputation for providing bad service to the product groups, there has been a failure of leadership by example.

Leadership may also fail to act because of its attachment to the status quo. In business, radical change in strategy or structure rarely happens without a change in leadership. Leaders may inaugurate change, but that change itself then becomes the new status quo.

Leadership is committed to the status quo for several reasons. The first is simple ego-involvement. If the current leaders were the architects of the status quo, then they will feel a special attachment to it as well as having a deep appreciation for its (original) functionality. The second reason is that

[7] The McKelvey (1976) Chaos Theorem established that absent a clearly dominant policy, a group using majority-rule voting can be led to any outcome by a sufficiently clever manipulation of the agenda.

[8] Many Marines value boot camp without liking it.

the status quo is known whereas new structures and strategies are not. As Burke pointed out in his critique of the French revolution, new and as yet untried systems generally have terrible and as yet unimagined flaws. The third and most subtle reason is that leadership, by its nature, involves commitment. The leadership of an organization must be the guarantor of explicit contracts, both internal and external. In particular, organizational members give energy and their own commitment to broad goals only to the extent that they perceive that the leadership itself has made a strong commitment to a particular set of priorities and ways of doing things. Major change requires a leader to repudiate prior commitments, thus lessening his or her ability to lead in the future. Hence, entrenched leadership is normally a source of inertia.

Embedded routines. The life functions of a business are its processes-its ways of doing things. Complex processes possess great inertia. The knowledge of how certain steps are performed may be tacit, no one may have a complete understanding of the process, and changing one aspect of a process may have significant unanticipated consequences on other parts of the organization. Finally, the various routines that make up the process take on the force of habit. From a purely economics perspective, organizational change only requires a change in incentives. However, the habitual patterns of work have an inertial force that can be much stronger than any practical incentives.

Process inertia arises not only through the attraction of the habitual, but also because change may mean doing things for which the organization has no process experience. If change simply requires substituting one known process for another, things are relatively straightforward. But taking the organization beyond its repertoire of routines is difficult. In general, one cannot ask an organization to do something which has not been reduced to a routine. If novel routines are required, then macro-routines for creating novelty, with all their slowness and cost, are required.

Collective action problems. Action can be blocked by a variety of collective action problems. The simplest is the first mover problem: if senior management has called for new initiatives, does it pay to be a first-mover? In many situations the incentives are clearly in favor of waiting to see how the first mover does. In such cases, the equilibrium is for no one to move at all. There are also analogous problems of free-riders that inhibit change even when first-movers have led the way.

The more complex collective action problem is that best described as cultural. [9] A dysfunctional culture may block change and itself be virtually impervious to alteration. Consider, for example, the simple question of cooperation (coordination) versus competition among departments. As a step towards a model, suppose that individuals in an organization meet in pairs and must each decide whether to cooperate (C) or compete (X). Each individual has information that the other does not and each controls some resources that the other does not. Cooperation means trying to act for the good of the company as a whole. In particular, it means making claims on the other person's resources that are justified by one's private information and accepting the other person's claims on one's own resources as valid. Competition means trying to act (covertly) for the good of the local unit. Concessions from the other are sought that will benefit one's own unit.

To make the game more concrete, example payoffs are shown in Figure 2. The net payoffs to

each player are symmetric and are simply weighted averages of the local and corporate payoffs. Figure 3 shows that if enough weight is placed on corporate payoffs, the equilibrium in the simple single-meeting game is **C**. If local payoffs dominate, the equilibrium is **X**. At intermediate levels of weighting, there are two pure-strategy equilibria: **C** and **X**. That is, there are two possible cultures and theory has little more to say about which will exist.

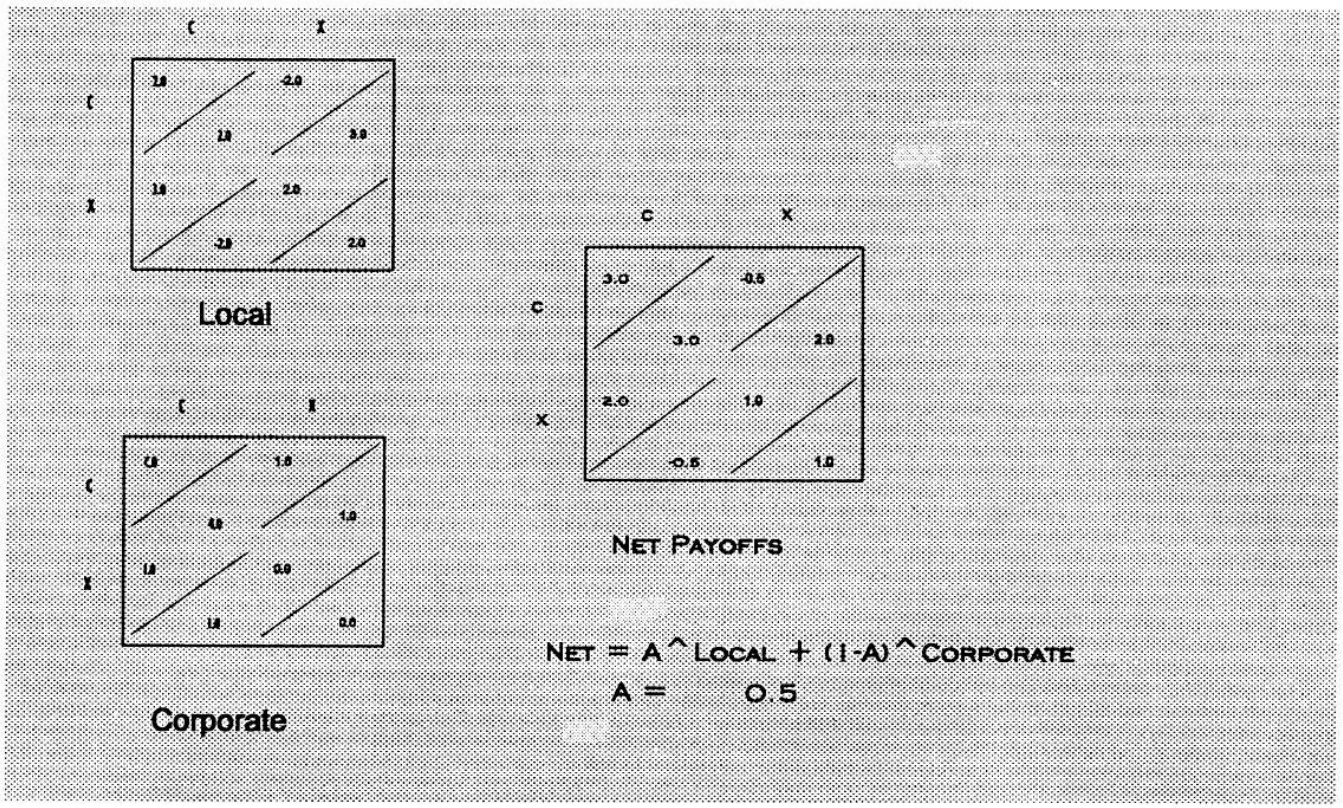

Figure 2 Payoffs in Cooperate vs. Compete Game

Suppose a firm had, over time, developed a culture having equilibrium **X**. How might management move it to **C**?[10] If a charismatic leader were able to convince the group that it was truly at **C**, then the equilibrium would indeed change. [11] Another strategy is to break off a small unit of the organization and focus resources and attention on changing its culture. Then, the old organization can be slowly recombined with the new culture in such a way as to preserve the new rather than overwhelm it with the old. A third strategy would be to merge with a firm having culture **C** and hope that the combination tilts in the right direction.

Because culture depends on mutual expectations, it is not easily changed. A culture that resists change or that does not fit the direction the firm needs to take can be an insurmountable source of inertia.

Capabilities gaps. The final action blockade is simply a gap or disconnect between the tasks that

[9] For a good discussion of the fundamental concept, see Ouchi and Wilkins (1985). For a technical view of the behavioral preconditions to social exchange, see Elster (1989).

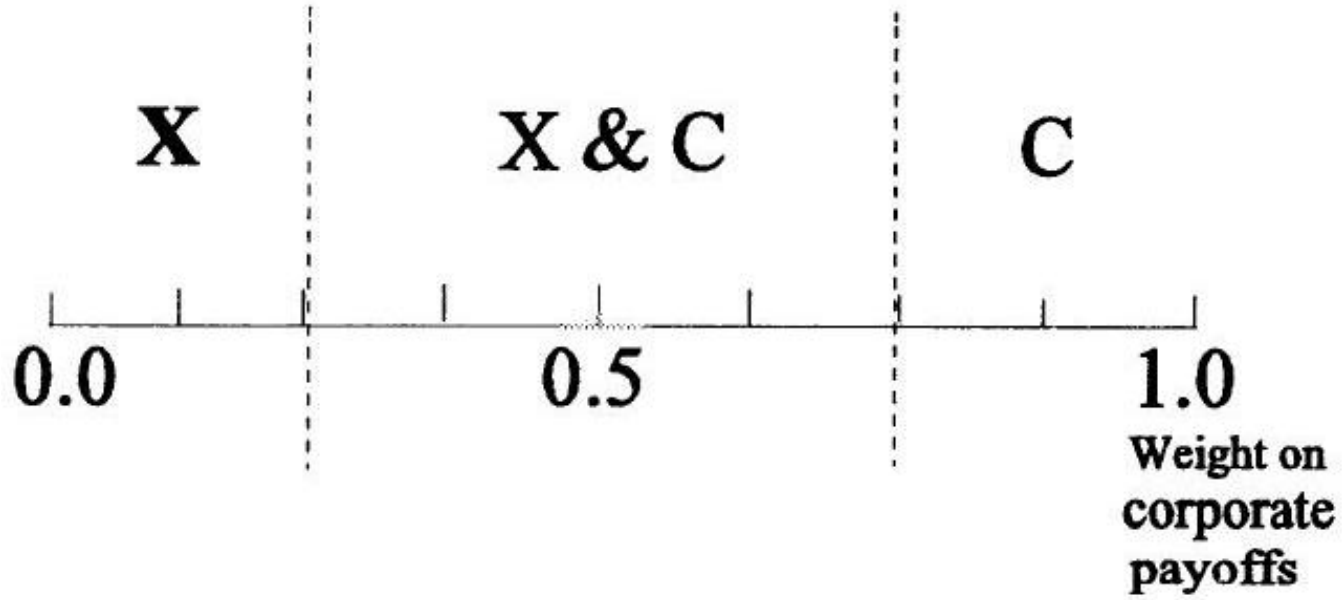

Figure 3 Equilibria vs. Payoff Weighting

need to be performed and the competencies and capabilities within the firm. Prahalad and Hamel (1990) have introduced the term "stretch" for the sense of tension between reality and aspiration, and have argued that healthy organizations are in a constant state of stretch. But too great a gap is discouraging and is more likely to inhibit than induce change.

The Process of Transformation

Transformation is the process of engendering a fundamental change in an organization with the goal of achieving to a dramatic improvement in performance. The fundamental change may involve strategic redirection, but always includes structural change and a dramatic alteration in the behavior of individuals. Transformation always involves overcoming large amounts of organizational inertia.

In certain exceptional cases inertia is greatly reduced by the existence of an already proven major invention or strategic innovation. Because the new business provides the motivation and the resources for change it takes the sting out of having to abandon old ways for new. Nourished by a steady stream of entrepreneurial rent, new structures and roles can flourish and develop as energy and attention are naturally drawn from mature or declining areas to the new richer terrain. Thus, for example, Intel's shift away from being a "memory chip" company was greatly facilitated by the fact that its microprocessor business was profitable and growing rapidly. Top management had only to coordinate the reallocation of resources from the old to the new business (Burgelman 1991).

Absent an innovation in hand, transformation contexts can be usefully divided into those in

[10] Assume incentives are not sufficient and complete monitoring is not feasible. [11] In this case one should play **C** if one believes that at least 60 percent of the others are playing **C**.

which operating efficiency is primary and those in which the adroit utilization or reallocation of resources is primary. Figure 4 is a guide to these contexts. The horizontal axis (capability) measures the relative efficiency of the firm at performing its tasks. The vertical axis (fit) measures the adequacy of those tasks in meeting product-market demands. The ideal firm is both capable and fit (adapted).

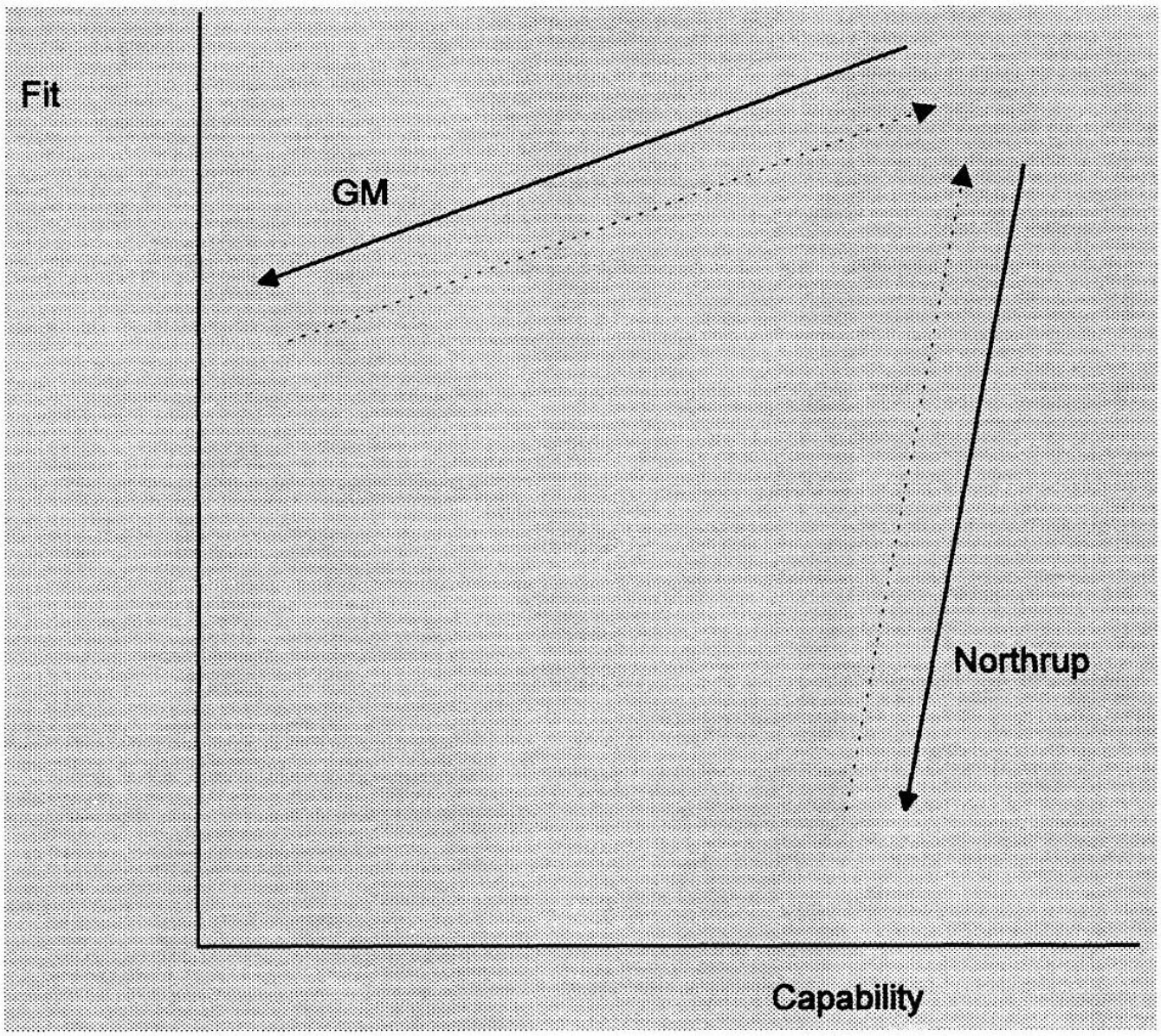

Figure 4 Change Contests

Corresponding to these two change contexts are two transformation tasks: recovery and renewal. Put simply, recovery is the process of regaining lost (relative) efficiency whereas renewal is the process of developing new skills and resources or of discovering new uses for extinct skills and resources.

Recovery is regaining efficiencies and skills that the firm used to have or would now possess if

it had followed the evolution of the industry leaders. Firms requiring recovery have been literally frozen in time by organizational inertia, failing to increase their efficiency as fast as the leaders. In some cases efficiency will have suffered an absolute decline. The rents financing such significant inefficiencies are the fruits of past strategic victories. A product patent may generate the rents that permit a company to live with sloppy manufacturing methods while close competitors are forced to improve their methods to survive in their unprotected markets. Or a company may enjoy a strong enough reputation with buyers, or be protected by large buyer switching costs, that it can survive years of declining efficiency, quality, and innovation.

The recent problems at General Motors, one could argue, stem from a decline in relative capability. The basic strategy of the firm, its overall product line, its distribution system, its essential product-technology, are not the problem. Thus, GM is a case of decline in capability rather than fit and the problems of inertia it faces are those associated with developing and learning new or improved capabilities. Its transformation challenge is therefore one of recovery.

By contrast, IBM remains a capable manufacturer of main-frame computers. Its recent problems arise from the growing lack of fit between its capabilities and the environment. Market demands are less for main-frames and more for very inexpensive PCs networked into work-groups. [12] IBM's fit with this new world is poor. Thus, the inertial problems IBM faces are quite different from those facing GM. Its management must deal with moving resources away from activities that are clearly being done well (on a technical basis) and which are valued for having been the wellsprings of the firm's past success. The transformation challenge faced by IBM is primarily one of renewal.

Modeling the Architecture of Organizational Competence

In order to discuss the specifics of transformation it is necessary to have a view as to the structure of a firm's competencies. The general viewpoint adopted here is that competencies are hierarchical in structure and exist in layers, the existence of a higher layer being dependent upon competent execution of lower layers.

An analogy to human skills may be helpful. Consider the performance of a helicopter pilot. The most fundamental skills (layer 1) are visual (sharpness, field of view, depth perception, etc.), basic motor coordination, and kinesthetic sense. Given layer 1 skills, the beginning pilot can learn specific tasks-taking off, landing, level flight, hovering, turning, and so on-that qualify as layer 2 skills. Given layer 2 skills, the pilot begins to integrate them into a fully coordinated competence at flying, where lower-level routines are called on unconsciously as needed and smoothly integrated. A layer 3 competent pilot will still need to work to accomplish layer 4-flying in close formation with other helicopters. [13] Clearly, layer 3 skills must be considerable and automatically available if layer 4 is to be attained. And the hierarchy continues-formation flying in bad weather, under combat

[12] Another firm clearly in this position is Northrop, the maker of the Stealth Bomber. Its capabilities at its historic mission are superb, but the mission appears to be obsolete.

conditions, etc.

Returning to the firm, one could obviously define skills in layer 1 as being basic manufacturing operations, selling activities, etc. Layer 2 might then be defined as the coordination of these skills into a coherent business. Layer 3 would then be the coordination among related businesses and layer 4 the adaptation of this pattern to local conditions throughout the world.

Whatever the complexity of reality in particular situations, the basic arguments I wish to advance can be made with a simple two-layer model. The model has the following characteristics:

1. Work performed by departments varies along four dimensions: task, method, waste, and routinization. a. Task is the production process being done-making plastic cups versus making paper cups. Major changes in strategy normally imply redesigned portfolios of tasks.

b. Method is the efficiency of the practices used to accomplish the task. A department's methods may be woefully inefficient or they may be state-of-the-art. c. Waste is the level of inefficiency, given the task and method. It represents the degree to which people work honestly and diligently and the degree to which supervision facilitates and encourages good work. d. Routinization is the degree to which the work has been practiced and can be smoothly performed. Note that poor methods and waste can nevertheless be accompanied by smoothly functioning routine. 2. Coordination skills (level 2) can only be developed when level 1 skills are routinized. 3. Coordination skills are specialized to the tasks and methods used by individual departments. 4. Individual departmental performance is measurable with much less error than individual contributions to coordinative activity. Level 1 department performance can be measured, compared across departments, benchmarked with comparable departments in other firms, and subjected to causal analyses. The overall effectiveness of Level 2 coordination can be measured (with error), but it is extremely difficult to assess the contributions of individual departments to the whole. 5. Coordination is costly to departments. It is work and it results in some degree of despecialization.

With regard to element 5, it is worth emphasizing that coordination is not costly just because it is work, but also because it interferes with routine. The most fundamental economic proposition about organization is the existence of gains to specialization. If there are gains to specialization, it means that it is most efficient for a department to concentrate on one task rather than spread its efforts among two or more tasks. So, to the extent that there are gains to specialization, there must be concomitant costs to coordination due to the despecialization it requires. [14]

Element 4 is a reminder that coordination is team production and that it is notoriously difficult to measure the individual contributions to team output. If monitoring of individual contributions is weak or inaccurate, then items 4 and 5 taken together create an incentive for free-riding by departments-rather than work at coordination they may simply stick to their knitting. To make more precise the idea that departmental work and coordinative work compete for a department manager's time and energy, the following conjecture is offered:

[13] In Vietnam U.S. pilots flew formations so tight that the rotors overlapped.

Conjecture 1 Increases (Decreases) in Departmental Incentive Intensity Reduce (Increase) Coordinative Activity.

Appendix A provides a demonstration of this conjecture in a simple setting.

The negative relationship between incentive intensity and coordinative activity is illustrated in Figure 5. As the requirement for coordination rises, the intensity of department incentives must be reduced. Conversely, reductions in the requirement for coordination permit increases in local department incentives. As the requirements for coordination drop below some critical level, say C_0, organizational governance is replaced by market governance and arm's-length transactions take place. Of course, the idea of a sharp dividing line is a gross simplification-some market relationships exist with fairly large levels of coordination, aided by bilateral dependence and repeated experience, and some interorganizational relationships have the sharp incentives and impersonality ordinarily ascribed to market transactions.

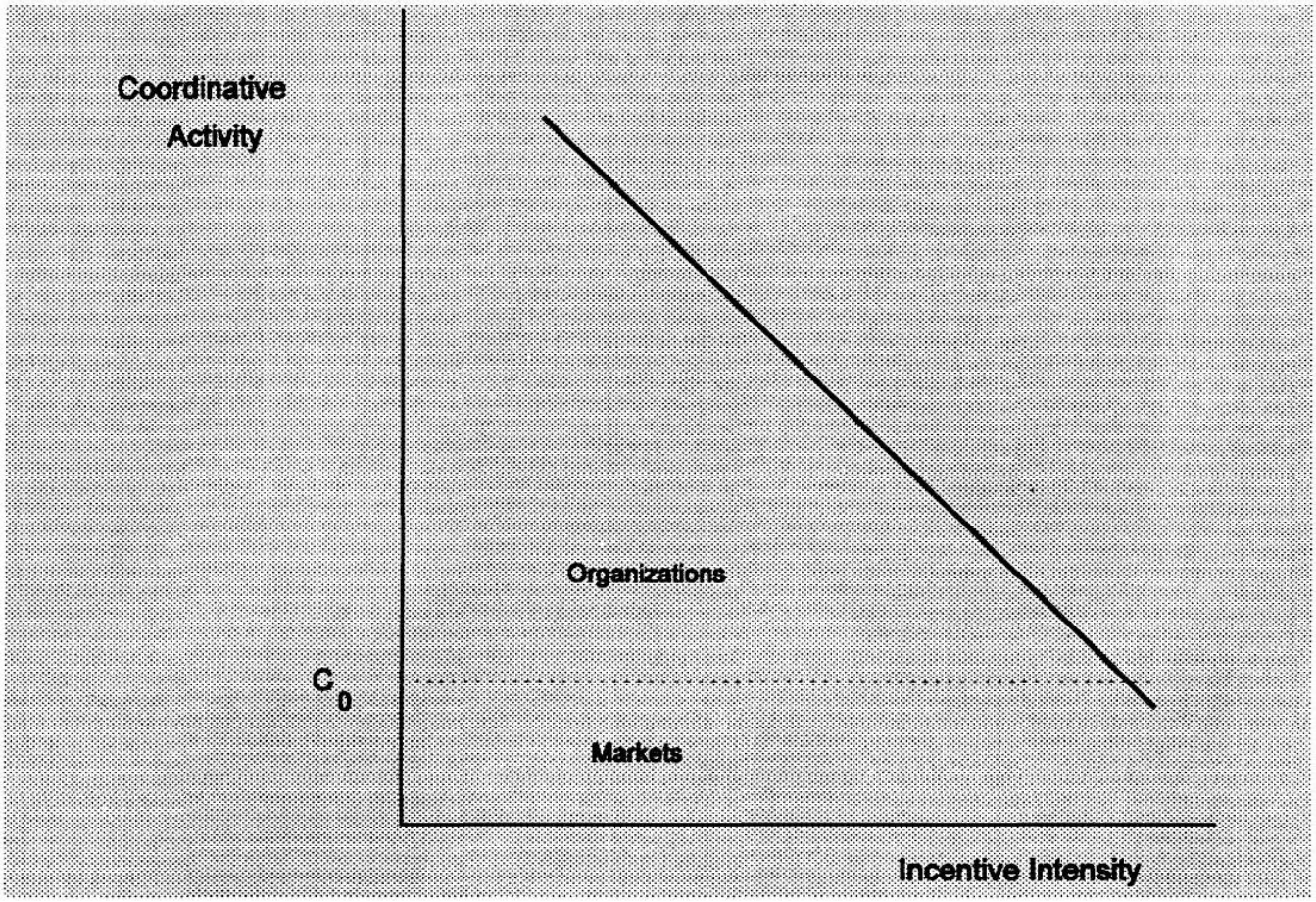

Figure 5 Coordinate Activity versus Incentive Activity

[14] For example, coordinating manufacturing to customer needs may mean interrupting production runs, using more set-ups, and increasing the variety of materials that must be carried in inventory. Of course it still may be optimal to bear these costs due to the extra value obtained through coordination.

Transaction cost economics focuses on the costs of reducing opportunism in bilateral exchange, where the specificity of the parties is induced by asset specificity. Such an analysis would see coordination as requiring investment in tangible or intangible capital specific to the other party and, therefore, as benefiting from non-market governance mechanisms. The argument advanced here does not rest on assumptions about asset specificity. Rather, if coordination is a team effort that is supplemental to basic department operations, then incentive intensity must be reduced below "marketplace" levels in order to divert effort away from department performance per se and towards coordinative activities. [15]

Note particularly that it is common for transaction cost analysts to see the reduction in incentive intensity accompanying internalization as an unavoidable cost to internalization. That is, with a full market interface between a buyer A and a supplier B, the supplier A has high powered incentives to produce efficiently. However, when A and B are departments within a single firm, incentives are unavoidably dulled. As Kreps (1990: 756) notes,

If A buys out B 's assets and employs B, then A is unable to match such strong incentives in the employment contract she gives to B. She has a difficult time monitoring the effort B expends, and she has an especially difficult time seeing how B expends whatever effort he does expend.

What is different in the analysis presented here is that it has firms existing to provide an escape from the high-powered incentives of the marketplace. Within the firm, where incentive intensity can be reduced by design, it is possible to generate more and richer coordinative activity than can be accomplished in markets. Thus, in contrast to transaction-cost theory, it is the gains to greater coordination that rationalize the firm.

Key Inertias

The description of the five frictions was intended to explore the range and depth of forces at work. Now, in order to facilitate reasoning within a simple model, it is useful to make some simplifying assumptions. Accordingly, I make the reasonable assumption that resistance to change is embodied in the following five inertias:

Cross-subsidy comforts from bundled businesses

Departmental politics rooted in self-interest

Embedded processes that link departments

Cultural inertia

Vested values in current methods and products

[15] Williamson (1985, Ch. 6) provides a good comparative view of the costs and benefits of both high and low-powered incentives both in firms and in markets.

Observation 1 Fragmentation and Increased Incentive Intensity Act to Reduce Each of the First Four Sources of Inertia.

Fragmentation means cutting the company into smaller departments, reducing interdependencies, measuring each department's performance and tying managerial pay and career potential to results. Fragmentation eliminates cross-subsidy comforts. It breaks political coalitions, asserts the center's legitimate authority over behavior, and compels attention to performance rather than discussion. It reduces the inertia of embedded (coordinative) processes by reducing their importance or by literally tearing them out. Fragmentation breaks the larger culture into smaller cultures, and cooperative behavior is more likely to emerge in smaller groups. Vested values are not strongly affected by fragmentation because it acts on behavior and attention rather than values or beliefs. Finally, fragmentation helps reveal the loci of department competencies, triggering key decisions on readjustments in the portfolio of departments and redesign of the overall strategy.

Of course, fragmentation can be very costly: it drastically reduces the amount of coordination within the firm. This, in turn, may sharply reduce the company's ability to design products, service customers, and perform other complex coordinative tasks. Furthermore, fragmentation may have little to do with the direction in which management would like to take the firm. Observation 1 does not claim that fragmentation is always a desirable direction. It merely notes that without fragmentation the journey may never begin.

Observation 2 Transformation Always Involves an Initial Reduction in the Amount of Coordination Among Departments.

If the transformation task is renewal, new departmental tasks must be defined and new methods discovered. Elements 2 and 3 of the competence model imply that changes in departmental methods imply replacing the present coordinative regime with a new one. That is, because coordination is specialized to task and method, new coordinative skills need be developed. This, in turn, implies that coordination will first be reduced, as the old system is scrapped and departments drive to adopt new methods.

If the transformation task is recovery, the need is to identify and remedy poor methods and waste. The practical approach to this issue involves some fragmentation, clearer measures, and increased incentive intensity. The focus on department performance will produce a reduction in coordinative activity (Conjecture 1). This leads directly to the following point:

Observation 3 the Cost of Fragmentation Is Reduced by the Necessity for Reductions in Coordination.

The cost of using fragmentation to cut inertia is the consequent loss of coordination. But Observation 2 assures us that coordination will be reduced anyway. Consequently, fragmentation becomes an even more attractive method for starting a transformation.

Observation 4 the Initial Stages of Transformation Will Be Accompanied by Increased Incentive Intensity.

This simply follows from Observation 3 and the necessity to motivate departmental action during the first stages of either renewal or recovery.

Transformation cannot, however, consist simply of fragmentation and increased incentive intensity. Business effectiveness also requires the close and subtle coordination among departments. But our model suggests that there is a necessary sequence of action in transformation. From the competence model (elements 2 and 3) we know that layer 2 cannot be built or developed until layer 1 is smoothly operating. Hence:

Observation 5 Rebuilding Coordination Among the Departments Must Await Task Redefinition, the Adjustment of Methods, the Elimination of Waste, and the Routinization of New Work Standards.

Coupling Observation 5 with Conjecture 1 leads to the final deduction about sequence:

Observation 6 Rebuilding Coordination Among the Departments Requires a Reduction in Incentive Intensity.

Transformation Profiles

According to the analysis just completed, the most direct way of reducing organizational inertia is to exploit the reduction in organizational cohesiveness that accompanies the first phases of transformation. In addition, the analysis points to a predictable sequence of events in organizational transformation. In the first phase, not directly discussed, managers within the firm become aware of the need for change and begin to formulate views as to appropriate new directions. In the second

phase, top management imposes structural fragmentation and increased incentive intensity. These moves have the effect of reducing coordinative activity, breaking some of the inertias that have impeded action, and focusing departmental attention on improving methods and eliminating waste. Once departmental performance has been improved, attention turns to rebuilding coordinative activity. To accomplish this, the incentive intensity must be reduced, else departments will have little reason to invest in difficult-tomeasure coordinative efforts. As coordination increases, best practices and other fruits of the central phase can be spread throughout the firm.

This sequence has a resemblance to that observed by Baden-Fuller and Stopford (1994). The middle two stages of their crescendo model are simplify and build, where the first involves removing outdated and unnecessary products, activities, and systems and the second is the construction, perhaps through trial and error, of new strengths, activities, and systems. However, the distinction emphasized here is between subunits with measurable outputs versus coordinative activities whereas their distinction is between the "backbone" elements of the business and the complex structure of less critical "weeds" that have grown up over time.

As the rebuilding of coordination requires the reduction in incentive intensity, it sets the stage for future problems. The benefits of increased coordination among departments should be clear, but it should also be clear that reduced incentive intensity invites another round of internal decay. Absent strong local performance incentives, waste can develop, poor methods can persist, and other inefficiencies appear. As we come to better understand this cycle of decay and renewal, we may someday know how to transform organizations without also planting the seeds of future decay.

Conclusions

If firms are not easily changed, there are important implications for strategy. The overwhelming evidence is that organizations possess considerable inertia, yet strategy content models, including the resource-based view of the firm, tend to sidestep this issue. A complete strategic theory of the firm must deal squarely with the issue of inertia.

Whereas the resource-based theory of the firm has had the advantage of using as a foil a corresponding model of the firm, there is no clear simple model of management or organizational process. Hence the study of inertia and transformation is complex and requires looking into a variety of disciplines.

In this article I have offered a number of ideas and conjectures garnered from economics, organizational sociology, cognitive psychology, political science, and general common sense. They have led to the elucidation of the five key frictions that impede change, a hierarchical view of the architecture of organizational competence, and a set of predictions about the time sequence of transformation phases.

Appendix a

Incentive Intensity and Coordination

In the discussion of competence architecture and transformation sequences, Conjecture 1 was offered, positing a negative relationship between incentive intensity (the strength of departmental performance incentives) and inter-department coordination. Looking closely at such an issue requires a dip into some agency theory. To accomplish this a modified version of Milgrom and Holmstrom's (1990) linear model will be analyzed. The modification is the introduction of multiple agents and payments based on team production.

A Linear Agency Formulation

Milgrom and Holmstrom (1990) build a principle-agent model in which the agent makes a choice of how much effort (or time) to spend on a number of tasks. The simplifications they introduce to permit analysis are linear wage rules, constant absolute risk aversion of the agent, independent normally distributed errors in measurement, and signals whose expected values are linear in the agent's effort. The model here is based upon theirs but adds the element of team production.

The agent's effort is divided between two tasks, with effort t_1 spent on the first and t_2 on the second. There are η agents and the principal observes each of their type-1 efforts individually with additive error ε_1 which is normally distributed with zero mean and variance σ_1^2. However, the principal can only observe the total type-2 effort τ put forward by all agents, again with additive error ε_2 distributed normally with mean zero and variance σ_1^2. All errors are independent. The agent's wage is

$$w = \alpha_1 \left(t_1 + \varepsilon_1\right) + \alpha_2 \left(\tau + \varepsilon_2\right) / n + \beta$$

Thus α_1 is the payment rate for local effort and α_2 is the payment rate for the average collective effort. The agent has constant absolute risk aversion r (that is, a utility function of the form $\mathrm{u(w)} = -\mathrm{e}^{-\mathrm{rw}}$). The agent's personal cost of effort is $\mathrm{C}\left(\mathrm{t}_1, \mathrm{t}_2\right)$, which is convex. With these specifications, the agent's certainty equivalent is

$$CE = \alpha_1 t_1 + \alpha_2 \tau / n - (r/2) \left[\alpha_1^2 \sigma_1^2 + \alpha_2^2 \sigma_2^2\right] - C + \beta$$

In solving the agent's maximization problem the question arises as to the value of $\mathrm{d}\tau/\mathrm{dt}_2$. If the agents are purely non-cooperative, then $\mathrm{d}\tau/\mathrm{dt}_2 = 1$, whereas fully cooperative behavior would imply $\mathrm{d}\tau/\mathrm{dt}_2 - n$. Assume that each agent possesses some zeal and increases his or her output by the fraction λ of the average increase in output per agent that can be attributed to others. Then

$$\frac{d\tau}{dt_2} = 1 + \lambda \frac{n-1}{n} \frac{d\tau}{dt_2}$$

so that

$$\frac{d\tau}{dt_2} = \frac{n}{\lambda + n(1-\lambda)}$$

For simplicity define $\mathrm{k} = \lambda + \eta(1-\lambda)$. When $\lambda = 0$ we have the non-cooperative solution ($\mathrm{k} = \eta$) and when $\lambda = 1$ we have the fully cooperative solution ($\mathrm{k} = 1$).

Let $\mathrm{C}_1 = \partial \mathrm{C}/\partial \mathrm{t_i}$. Then the agent's maximization problem is solved (as long as $\mathrm{t_i} > 0$) by setting

$$C_1 = \alpha_1$$

and

$$kC_2 = \alpha_2.$$

Define $\mathrm{C_{ij}} = \partial^2 \mathrm{C}/\partial \mathrm{t_i} \partial \mathrm{t_j}$ and take partial derivatives of (4) and (5) to give

$$\partial \alpha_1 / \partial t_i = C_{1i}$$

and

$$\partial \alpha_2 / \partial t_i = kC_{2i}.$$

Using the Inverse Function Theorem we have $\begin{pmatrix} \partial t_1/\partial \alpha_1 & \partial t_1/\partial \alpha_2 \\ \partial t_2/\partial \alpha_1 & \partial t_2/\partial \alpha_2 \end{pmatrix} = \begin{pmatrix} C_{22} & -C_{12}/k \\ -C_{12} & C_{11}/k \end{pmatrix} / \left[C_{11}C22 - C_{12}^2\right].$

The expected state of affairs is for $\mathrm{C_{ij}} > 0$. That is, more effort on some task should make additional effort on it or some other task even more costly. In addition, we expect $\partial \mathrm{t}_1/\partial \alpha_1 > 0$ and $\partial \mathrm{t}_2/\partial \alpha_2 > 0$ because a larger incentive placed on task i should call forth more effort on task i. Consequently, the denominator of the right-hand matrix must be positive in a well-behaved specification. The sought-after result is thus immediate: $\partial \mathrm{t}_2/\partial \alpha_1 < 0$. Increased local incentive intensity (α_1) reduces effort devoted towards coordination (t_2). It can also be observed that this connection does not depend upon the level of cooperation (k).

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